

AI-Driven Customer Intelligence System for Strategic Business Decision Making

Capstone Project – Unsupervised Machine Learning

Domain: Banking Customer Segmentation

Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Platform: VS Code

1. Introduction

Organizations today collect large volumes of customer data but often lack labeled information to understand customer behavior effectively. Without clear customer segmentation, businesses struggle to identify high-value customers, predict churn, and optimize marketing strategies.

This project focuses on building an **AI-driven customer intelligence system** using **unsupervised machine learning techniques** to discover hidden customer segments and convert them into actionable business insights.

2. Problem Statement

A banking organization possesses extensive customer data but does not know:

- Who their most valuable customers are
- Which customers are at risk of churn
- Which segments respond better to marketing campaigns
- How to optimize customer engagement strategies

Since no labeled data is available, **unsupervised learning** techniques are required to identify meaningful customer segments.

3. Dataset Description

- **Dataset Name:** Bank Marketing Dataset
- **Source:** Kaggle
- **Number of Records:** ~5,000+ customers
- **Data Type:** Real-world banking customer data

Key Features:

- Age
- Job
- Marital Status
- Education
- Account Balance
- Call Duration
- Campaign Interaction Count
- Previous Contact History

The dataset contains demographic, financial, and behavioral attributes suitable for customer segmentation.

4. Project Methodology

The project follows the complete machine learning lifecycle:

- Data Understanding
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Clustering Model Implementation
- Cluster Optimization & Evaluation
- Dimensionality Reduction & Visualization

- Cluster Interpretation & Business Insights

5. Data Preprocessing

Data preprocessing was performed to ensure data quality and consistency:

- **Missing Value Handling:**
Numerical features were filled using median values, while categorical features were filled using mode.
- **Outlier Detection & Treatment:**
Outliers were detected using boxplots and treated using the IQR capping method to retain customer behavior information.

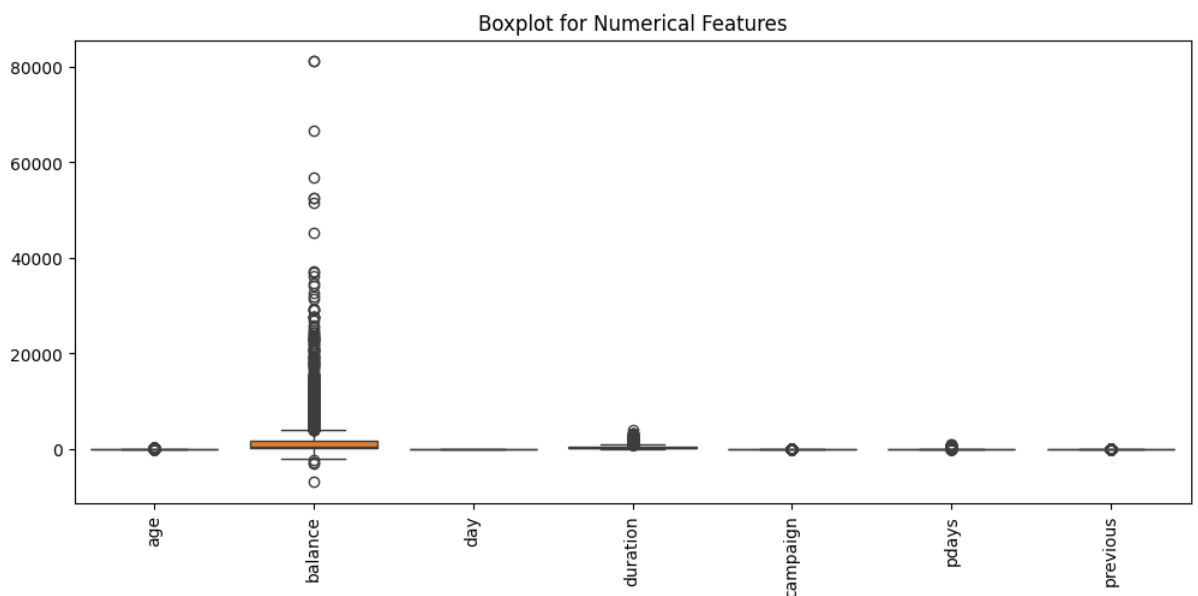


Figure 1: Boxplot showing distribution and outlier detection for numerical features.

- **Categorical Encoding:**
Label encoding was applied to categorical variables.
- **Feature Scaling:**
StandardScaler was used to normalize numerical features, ensuring equal contribution during clustering.

6. Exploratory Data Analysis (EDA)

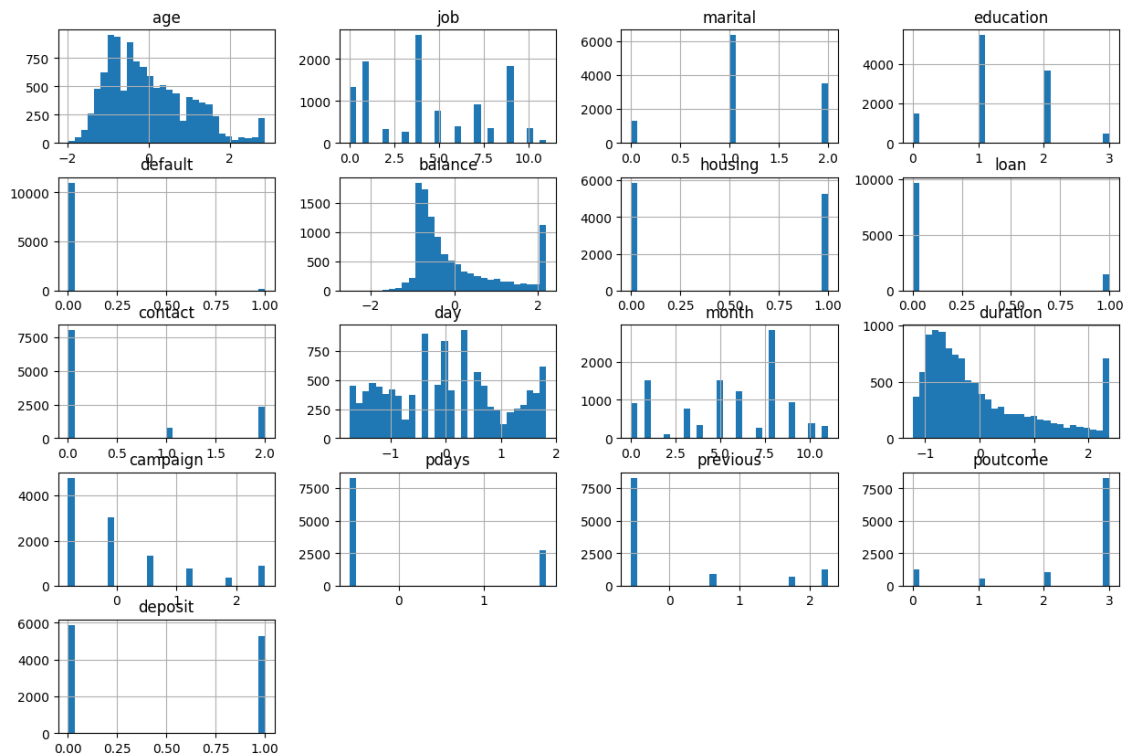


Figure 2: Univariate distribution of customer features.

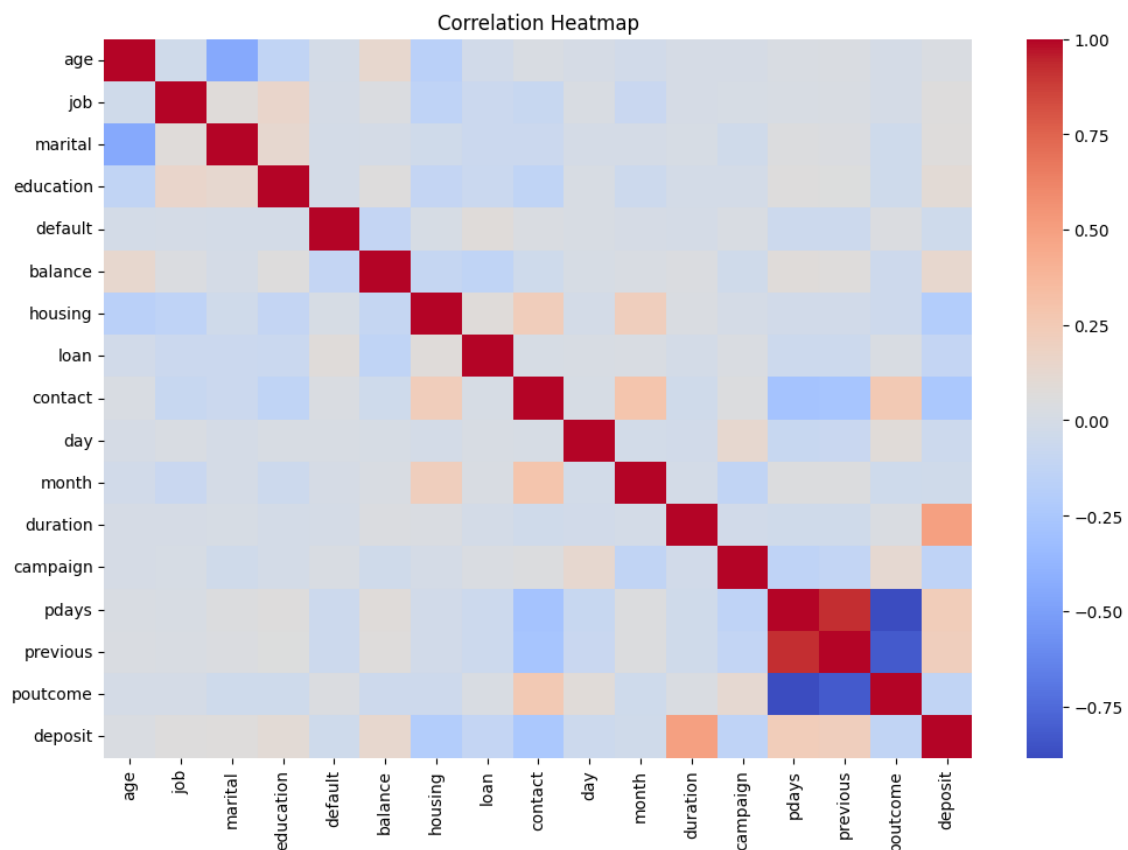


Figure 3: Correlation heatmap showing relationships among features.

EDA was conducted to understand data distribution and patterns:

- Univariate analysis revealed age to be normally distributed, while balance and duration were right-skewed.
- Correlation analysis highlighted relationships between call duration, campaign interactions, and customer engagement.
- Campaign-related features showed that most customers were contacted a limited number of times.

EDA provided insights necessary for effective feature engineering and clustering.

7. Feature Engineering

To improve clustering quality, additional behavioral and derived features were created:

- **Interaction Score:** Combined campaign and previous contact counts
- **Average Contact Intensity:** Call duration per campaign
- **Balance per Contact:** Financial value relative to marketing effort
- **Recency Score:** Proxy derived from previous contact days

Feature engineering enhanced behavioral representation and segmentation accuracy

8. Clustering Algorithms Implemented

The following unsupervised learning algorithms were implemented and compared:

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Model (GMM)

Multiple algorithms were used to ensure robustness and meet industry-level requirements.

9. Cluster Optimization & Evaluation

To determine the optimal number of clusters, the following methods were applied:

- Elbow Method

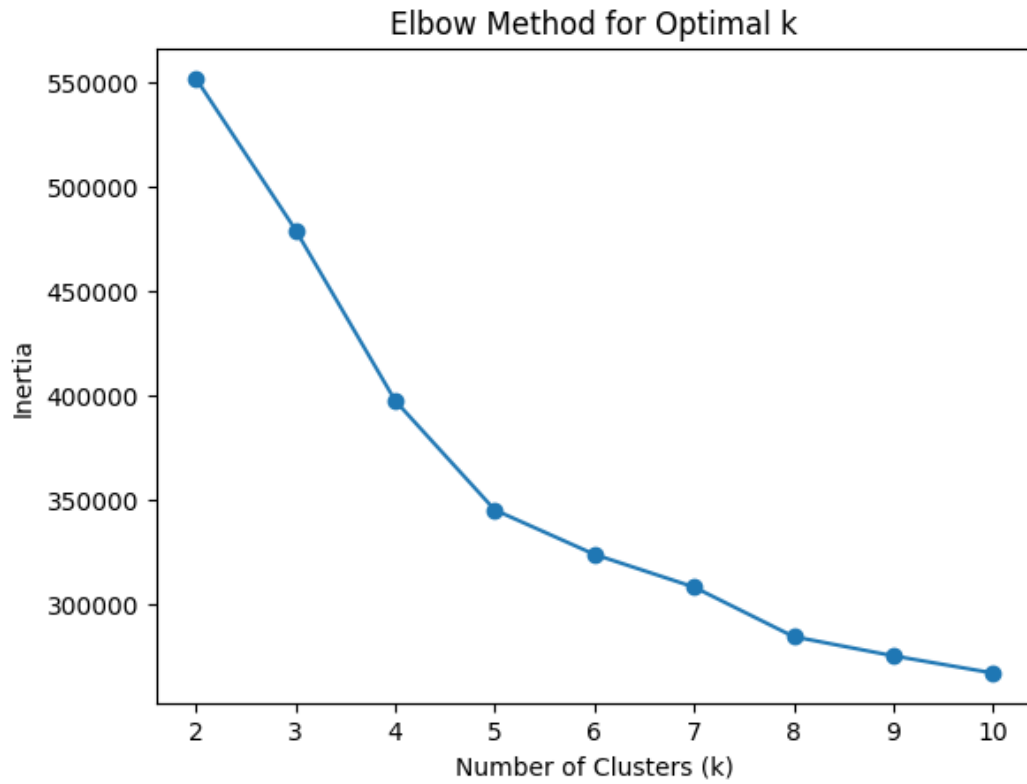


Figure 4: Elbow method used to determine optimal cluster count.

- Silhouette Score

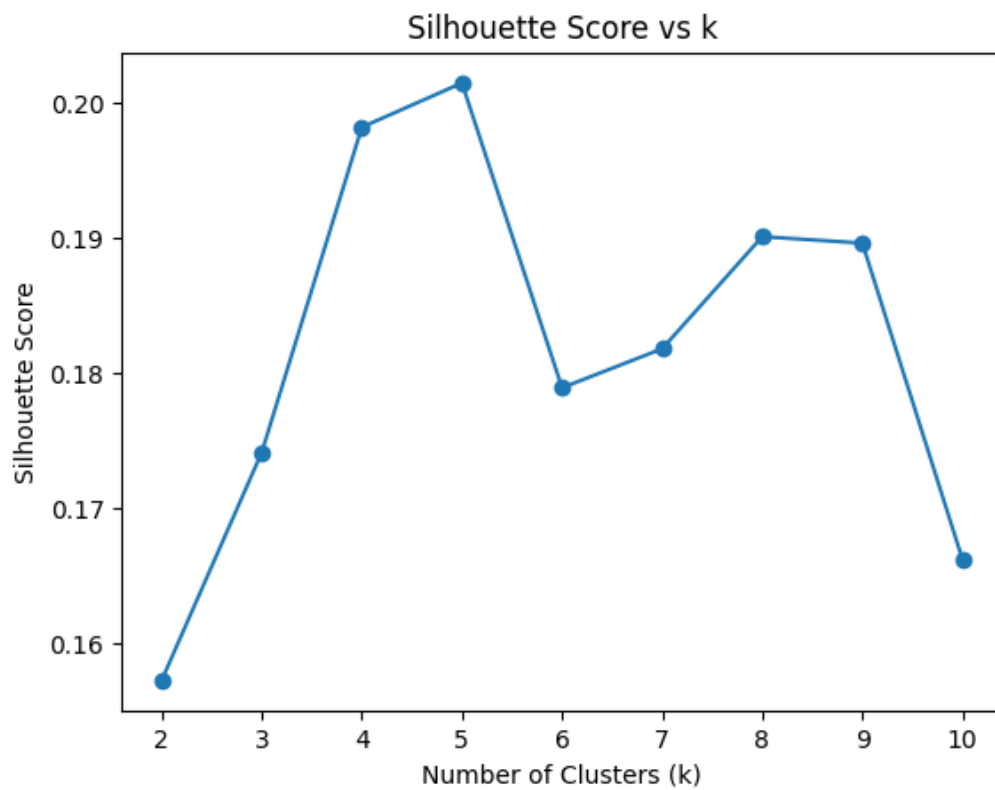
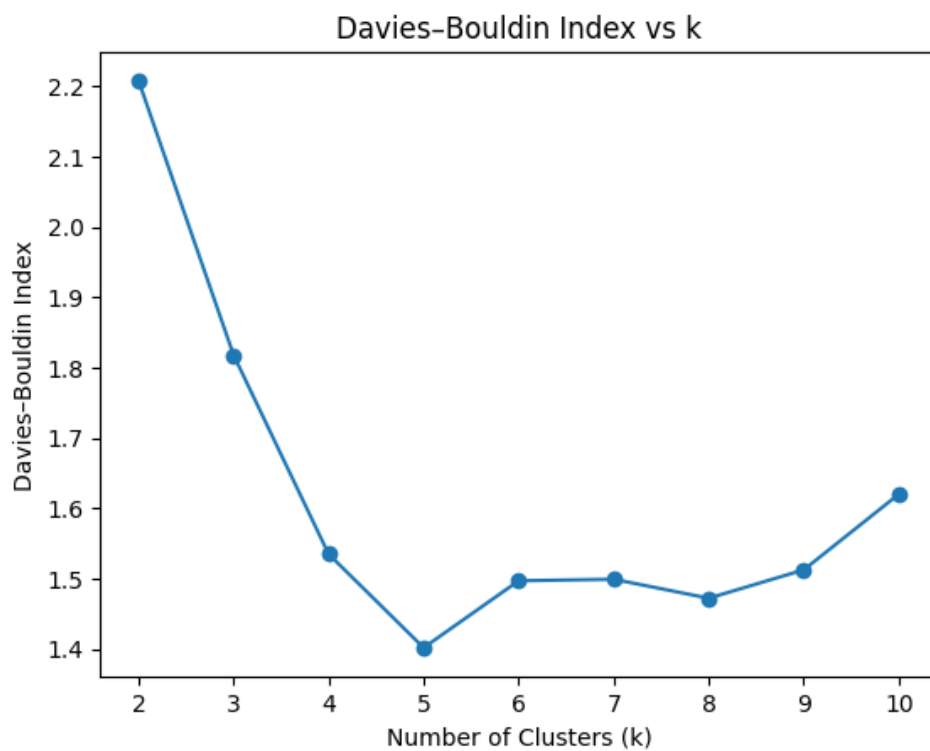


Figure 5: Silhouette score comparison across different k values.

- Davies–Bouldin Index



Based on all three metrics, 4 clusters were selected as the optimal segmentation.

10. Dimensionality Reduction & Visualization

- **Principal Component Analysis (PCA)** was applied to reduce dimensionality and visualize clusters in 2D space.
- **t-SNE** was additionally used to capture non-linear cluster separation.

Visualization confirmed clear cluster separation, validating the clustering approach.



Figure 6: PCA visualization showing separation of customer segments.

11. Business Insights & Recommendations

Cluster 0 – High-Value Engaged Customers

- High balance and strong engagement
- Recommended for premium services and loyalty programs

Cluster 1 – Low-Engagement At-Risk Customers

- Low balance and weak engagement
- Requires retention-focused strategies

Cluster 2 – High Contact but Low Conversion Customers

- Frequent marketing contact with limited returns
- Marketing strategy optimization required

Cluster 3 – Stable Moderate-Value Customers

- Consistent engagement and average value
- Suitable for cross-selling opportunities

Key Business Outcomes

- Highest revenue generated by Cluster 0
- Cluster 1 identified as churn-prone
- Targeted marketing reduces costs and improves ROI

12. Conclusion

This project demonstrates the effectiveness of unsupervised machine learning in deriving valuable customer insights without labeled data. The developed system enables data-driven marketing, improved retention strategies, and strategic business decision-making.

13. Future Enhancements

- Autoencoder-based clustering
- Time-based customer segmentation
- Customer Lifetime Value (CLV) modeling
- Hybrid clustering techniques